

Solution Requirements

Date	26 June 2026
Team ID	SWTID-2026-9014
Project Name	AdForge AI: AI-Powered Advertisement Copy Generator
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Groq API key is securely managed using environment variables (.env).</i>	High
2	Authorization levels	<i>Single user access; no separate admin or user roles.</i>	Low
3	External interfaces	<i>Integrates with the Groq Cloud API to generate AI-powered advertisement content.</i>	High

4	Transactions processing	<i>Accepts campaign details, processes AI requests, and returns generated advertisements without storing user data.</i>	High
5	Reporting	<i>Displays generated ad copy, slogans, performance tips, and refined content on the dashboard.</i>	Medium
6	Business rules, etc.	<i>Generate platform-specific ads based on selected platform, tone, audience, and campaign goal.</i>	High
7	Compliance to laws or regulations	<i>Protect API credentials, sanitize user inputs, and avoid generating harmful or inappropriate content.</i>	High
8	<i>Other</i>	<i>Allow users to refine existing advertisement copy and download generated results.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>Generate advertisement content quickly.</i>	<i>Response within 5–10 seconds</i>
2	Scalability	<i>Support multiple users requesting AI-generated ads simultaneously.</i>	<i>Support 100+ concurrent requests</i>
3	Security & Data Privacy	<i>Secure API key management and input sanitization.</i>	<i>API key stored in .env and all inputs validated</i>
4	Reliability & Availability	<i>Ensure stable application performance during normal usage.</i>	<i>99% uptime</i>
5	Usability & Accessibility	<i>Responsive, user-friendly interface accessible on desktop and mobile browsers.</i>	<i>Fully responsive UI</i>
6	Other	<i>Maintainable and modular Flask code for future enhancements.</i>	<i>Clean, reusable, and well-documented code</i>